

The Programmable Trust

Understanding Blockchain

Instant value transfer between strangers, without intermediaries.

Trust guaranteed by math and networks, not by institutions.

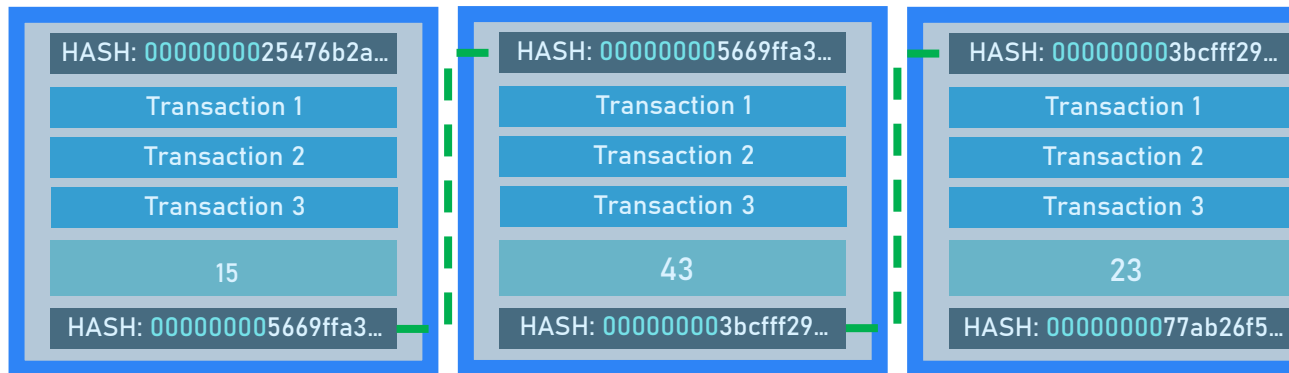
The most important technological shift of our generation.



Your Roadmap to Understand Blockchain

Seven key concepts to master:

- 1 Centralization vs. Decentralization
- 2 The Blockchain Architecture
- 3 Trust Without Intermediaries
- 4 Proof of Work and Mining
- 5 Keys Public and Private
- 6 Real World Applications
- 7 Challenges



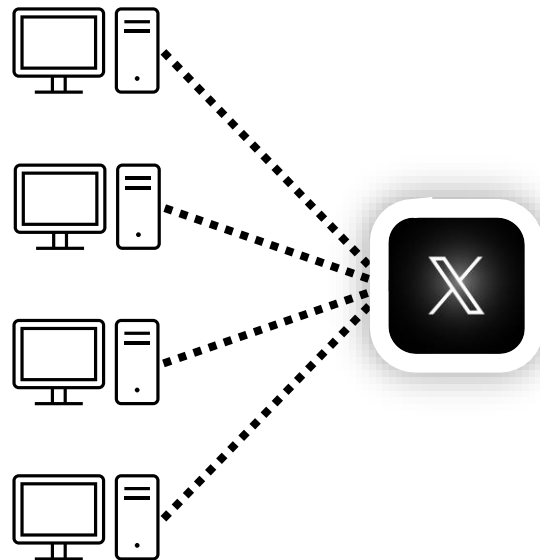
The Fragility of Centralized Architecture

Single Point of Failure: If the central server fails, the entire network stops

Control Concentrated: All power and decision-making in one entity

Trust Required: Must trust the central authority completely

Limited Transparency: Users cannot verify what happens behind the scenes



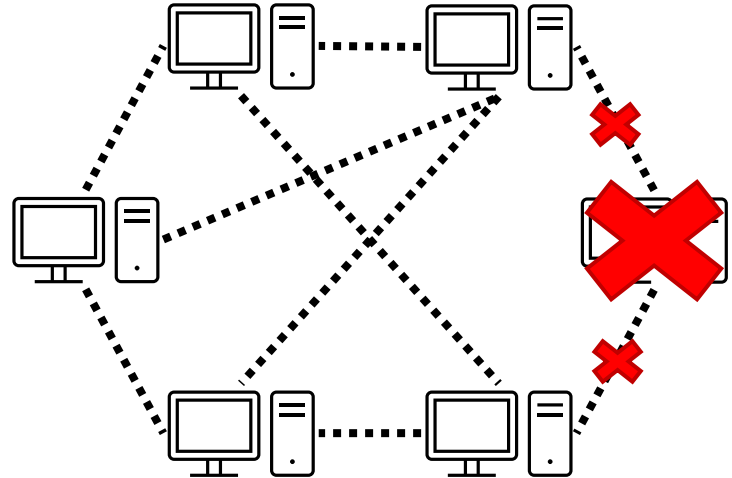
Peer-to-Peer (P2P) Networks: A Better Way

Every Node is Equal: No central server, all participants are peers

Distributed Storage: Each node stores data and processes transactions

Resilience: If one node fails, the network continues operating

The Challenge: How do we reach consensus without a central authority?



Blockchain: A Distributed, Immutable Ledger

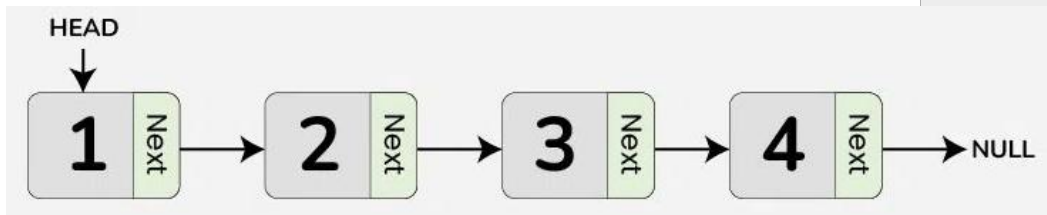
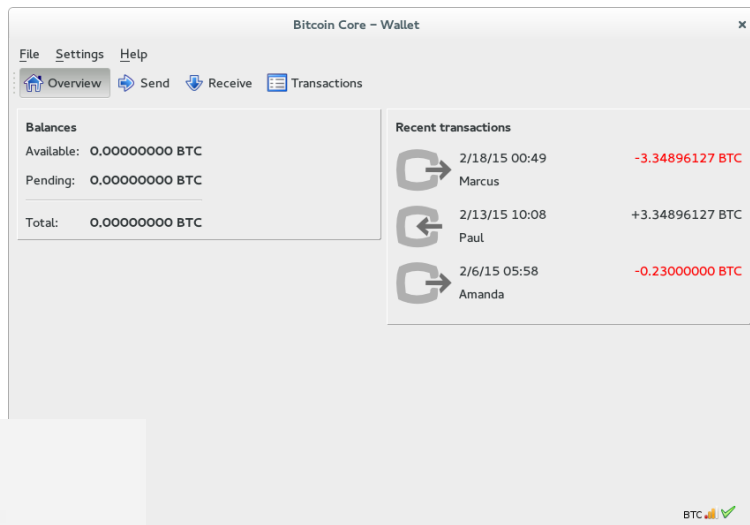
A distributed **database** replicated across thousands of computers, where data can only be added, never modified or deleted.

Distributed: Replicated across thousands of computers, not controlled by a single entity

Append-Only: Data can only be added; existing entries cannot be modified or deleted

Transparent: All transactions are visible and verifiable by all participants

Immutable: Once recorded, data is permanent and cannot be erased without notice



Each block contains transactions and references the previous block's hash, creating an unbreakable chain

Solving the Time Synchronization Problem

How blocks create temporal order in distributed networks

1

The Problem

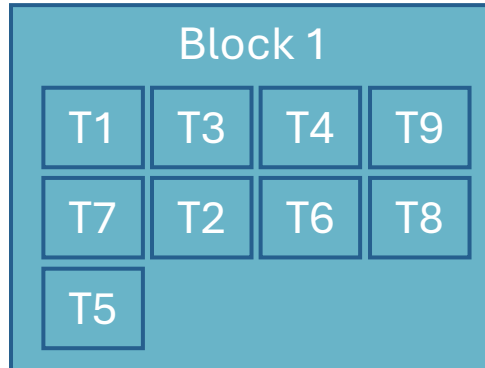
Transactions arrive at different times across the network. Clock synchronization is impossible.



2

The Solution

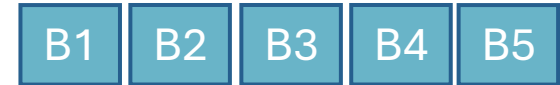
Group transactions into blocks. All transactions in a block are considered simultaneous.



3

The Result

Blocks create a chronological sequence. The blockchain orders events reliably.



Anatomy of a Block: The Building Blocks of Trust

Block Header

Version: 0x00000001

Difficulty(nBits): 1

Previous Hash: 00000000839a8e6886ab5951d76f4114
75428afc90947ee320161bbf18eb6048

Merkle Root: 9b0fc92260312ce44e74ef369f5c66bbb85848f2e
ddd5a7a1cde251e54ccfd5

Timestamp: 2009-01-09 03:55:44

Nonce: 0x61bdd208

Block Body

Transaction 1: Carlos -> Elon Musk: 2 **BTC**

Transaction 2: Álvaro -> Pedro: 50 **BTC**

Transaction 3: Satoshi -> Nakamoto: 1000 **BTC**

Transaction X: XXXXX -> XXXXXXXXX: XXX **BTC**

Block Identity

Block Hash: 000000006a625f06636b8bb6ac7b960a
8d03705d1ace08b1a19da3fdcc99ddb

- 1 Previous Block's Hash**
The cryptographic link to the previous block. This creates the chain. If any data in the previous block changes, its hash changes, breaking the link.
- 2 Merkle Root**
A single hash that summarizes all transactions in this block. It's like a fingerprint of all transactions, enabling efficient verification.
- 3 Timestamp**
Records when the block was created, providing chronological order to the ledger.
- 4 Nonce**
A number used in mining. Miners change this to find a hash that meets the difficulty target.
- 5 Block Identity**
The hash of the header. Change any single bit of data, and the entire hash changes, making tampering instantly detectable.

Cryptographic Hashing: The Foundation of Security

Key Properties of Hashing

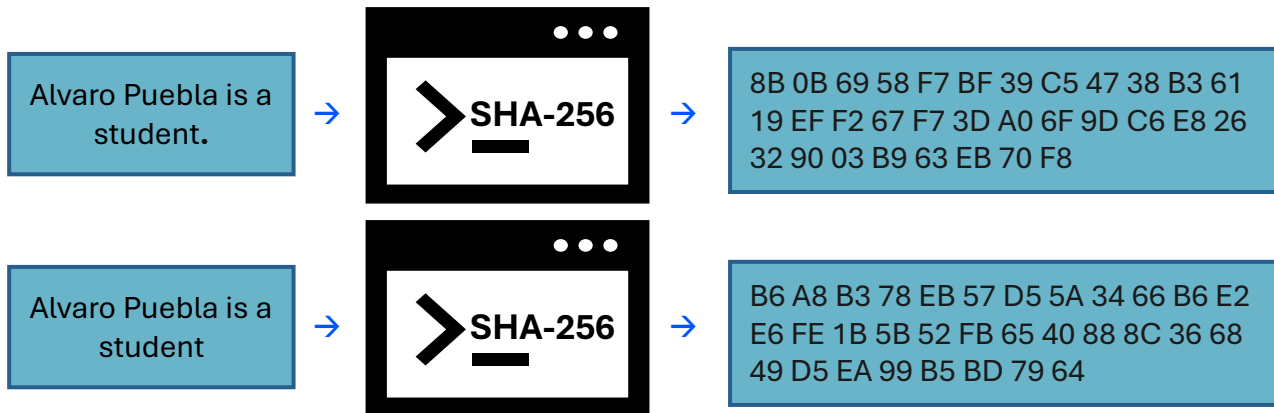
Deterministic: Same input always produces same output. Hash the same data twice, get identical results.

Avalanche Effect: Tiny change in input produces completely different output. Change one bit, and the entire hash changes.

One-Way Function: Cannot reverse the hash to get the original input. You can only verify by hashing again.

Collision-Resistant: Virtually impossible to find two different inputs that produce the same hash.

Practical Example SHA-256 bits:



The Sybil Attack: Why Consensus is Hard

The fundamental challenge of voting without a central authority

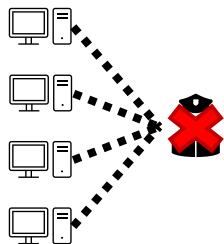
1

The Problem

How do we vote?

In a decentralized network, there's no central authority to count votes or decide which block is valid.

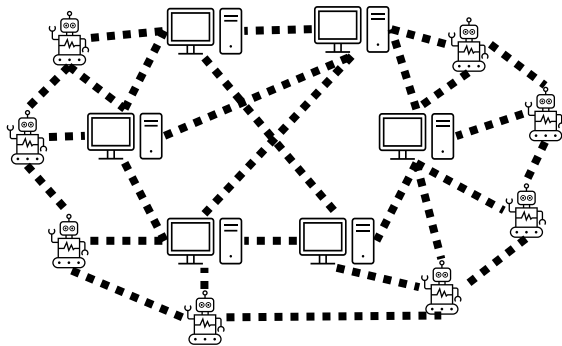
We need all participants to reach consensus.



2

The Attack Creating Fake Identities

One attacker can create thousands of fake nodes (identities: IP, account, person?) to manipulate the vote. This is called a Sybil Attack.



3

The Real Solution

Make participation costly. Require each participant to expend computational resources. This is where **Proof of Work** comes in.



Proof of Work: Earning the Right to Mine

How miners compete to secure the network

1 The Challenge

Find a (random number) that makes the block's hash start with a specific number of zeros (lower than a given target).

Input:

Block header



SHA-256 Hash Function



Output: 000000abcd1234...
(Must start with N zeros)

Target:

HASH of the block < Difficulty

2 The Process

Trial and Error: Miners test billions of Nonce values until one produces a valid hash.

Nonce = 1 → Hash = 3a7f2c9e...
Nonce = 2 → Hash = 7b2a1f4e...
Nonce = 3 → Hash = 1c8d5e2b...
...
Nonce = 2,147,483,648 →
Hash = 000000abcd... ✓ Found!

Difficulty Adjustment:

Automatically adjusts to maintain consistent block creation time (~10 min for Bitcoin)

3 The Reward

The miner who finds the solution first wins the right to add their block to the chain.

Reward = Newly Created BTC
+ Transaction Fees

Incentive: Miners are motivated to work on the main chain, securing the network

Cost: Massive computational power and electricity required, making attacks economically prohibitive



The Longest Chain is the Truth

How the network reaches consensus on the canonical ledger

Chain A:     ✓ **CANONICAL**

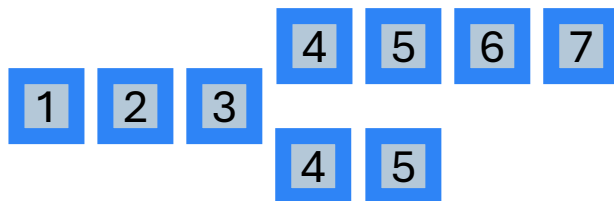
4 blocks | More confirmations | Most computational work

Chain B:   

3 blocks | Orphaned | Abandoned by network

Fork Resolution: When two chains compete, nodes follow both temporarily. As Chain A grows longer, nodes abandon Chain B. Miners working on Chain B lose their rewards.

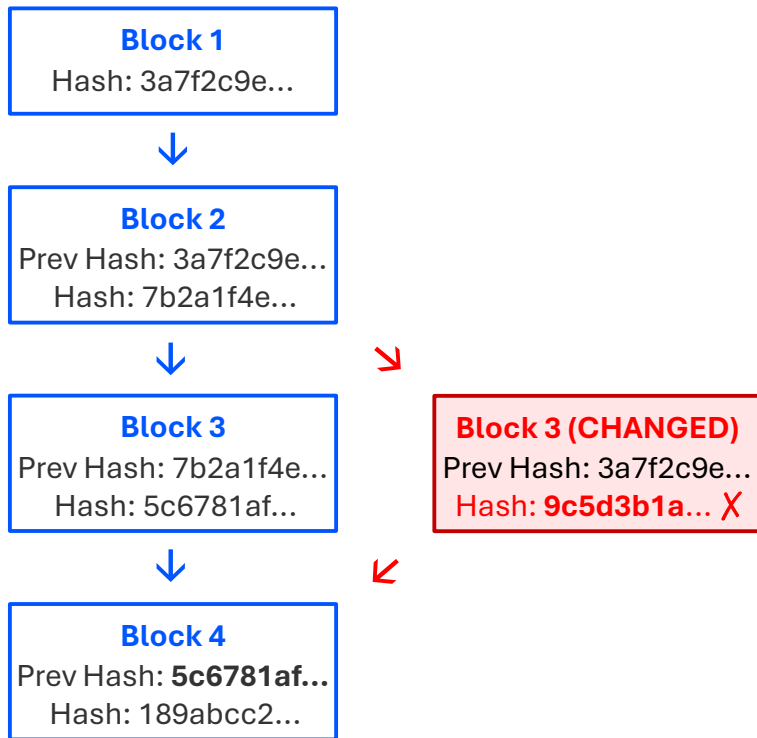
Hard Fork and Soft Fork



Consensus Rule

- 1 Longest Chain Wins**
The network always accepts the longest valid chain as the true version of the ledger.
- 2 Computational Work**
The longest chain represents the most computational work performed by the network.
- 3 Finality**
After several confirmations (e.g., 6 blocks), transactions become essentially irreversible.
- 4 Incentive Alignment**
Miners are incentivized to work on the main chain because that's where rewards are valid.

Why Blockchain is Secure: The Domino Effect



Why Tampering is Impossible

- 1 Chaining**
Each block contains the hash of the previous block, creating an unbreakable chain.
- 2 Avalanche Effect**
Change one byte in Block 3, and its hash changes completely, breaking the link to Block 4.
- 3 Cascade**
The attacker must recalculate the hash for Block 4, which changes its hash, breaking the link to Block 5, and so on.
- 4 The Race**
The **attacker must outpace the entire network** adding new blocks. Computationally impossible.

The 51% Attack: Why It's Impractical

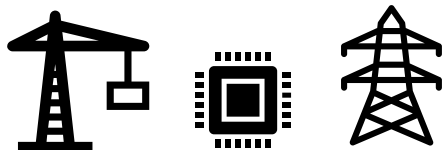
Economic analysis of attacking a major blockchain network

1 What and why

An attacker gains control of the network's total computational power (hash rate).

With the full control of the network, they can:

- Create the longest fork
- Reverse recent transactions (double-spend)
- Prevent new transactions from being confirmed



2 Cost

The attacker would **need resources equivalent to more than half of the network's total spending.**

Hardware -> Millions of miners
Electricity -> **Gigawatts** of continuous power every day
Infrastructure -> Warehouses / Cooling systems / Maintenance

A realistically executed 51% attack on Bitcoin would likely cost **tens of billions of US dollars:**
30,000,000,000 \$ is what **1.2 million** Polish people generate in 1 year

3 Why is impossible

Economic Irrationality:

You'd spend billions to steal millions.

Immediate Detection

The network would instantly detect and hard fork to change the consensus rules to exclude the attacker.

Network Response

The community would hard fork to change the consensus rules, rendering the attack useless.

Network Growth

As the network grows, the cost of a 51% attack increases **exponentially.**

The keys: Private and Public

What proofs that you own your assets

Private Key

- A secret string of characters that proves ownership of cryptocurrency.
- Must be kept secure because anyone with it can access your funds.
- Used to sign transactions and unlock your assets.

If stolen or lost -> **Assets will be lost forever!!!**



Bitcoin Example:

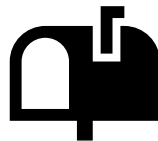
Curve (secp256k1 $\rightarrow y^2 = x^3 + 7 \pmod{p}$)

Where $p = 2^{256} - 2^{32} - 2^9 - 2^8 - 2^7 - 2^6 - 2^4 - 1$

Then given the standard $G(x,y)=(79BE667E..., 483ADA77...)$ and its order $n \approx 1.158 * 10^{77} \Rightarrow n * G = O$ ("zero" of elliptic-curve)

Your private key will be a random number that you choose in the range: $1 \leq PrivateKey < n$

Public Key



- A derived string from the private key that can be shared openly.
- Acts as your blockchain address, allowing transactions to be verified without revealing your private key.
- Used by others to send you cryptocurrency.

This is like your mailbox, anyone can send you a letter

To calculate your public you just need to calculate:

$$PublicKey = PrivateKey * G$$



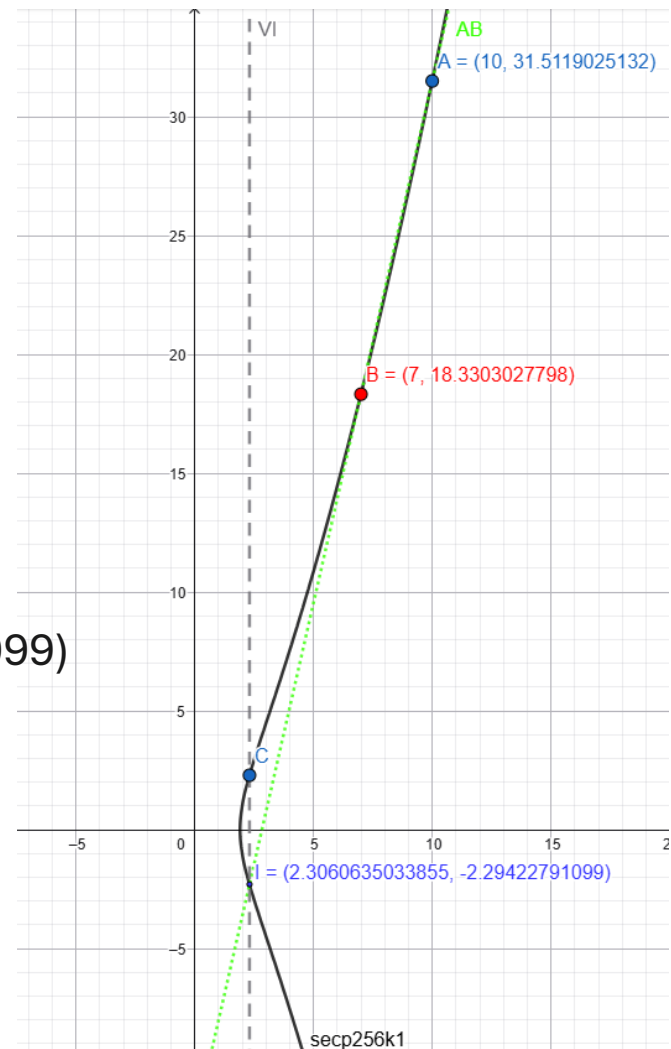
$$\text{PublicKey} = \text{PrivatKey} * G :$$

Given some example parameters:

- Curve $y^2 = x^3 + 7$
- $A(\text{privateKey}) = (10, 31.5119025132)$
- $B(G) = (7, 18.3303027798)$

You need to calculate: $C = A+B$

- $I = (2.3060635033855, -2.29422791099)$
- $C(\text{publicKey}) = (2.3060635033855, 2.29422791099)$



Blockchain in the Real World

Practical applications transforming industries



Supply Chain Transparency

Track products from origin to consumer with immutable records.

Verify authenticity and prevent counterfeits

Reduce fraud and improve efficiency



Financial Services

Enable instant settlements and cross-border payments without intermediaries.

Lower transaction costs

Financial inclusion for unbanked populations



Digital Identity

Self-sovereign identity controlled by individuals, not institutions.

Secure, portable identity credentials

Privacy and control over personal data



Logistics & Tracking

Real-time tracking of shipments with transparent, verifiable records.

Reduce delays and improve transparency

Enable automated payment



Voting Systems

Transparent, tamper-proof voting that's auditable and verifiable.

Prevent fraud and ensure election integrity

Increase voter participation and trust



Healthcare Records

Secure, interoperable medical records accessible to authorized providers.

Patient control over health data
Improved care coordination and research

The Challenges: Energy, Regulation, and Scalability

Energy Consumption

Proof of Work requires significant computational power, consuming substantial electricity

Bitcoin alone uses ~150 TWh/year
(comparable to **Argentina**)

Environmental impact concerns



Multiple approaches to reduce energy consumption while maintaining security.

- **Proof of Stake:** Ethereum PoW -> PoS, reducing by 99.95%
- **Renewable Energy:** Mining powered by renewables
- **Layer 2 Solutions:** Reduce on-chain transactions

In Progress ...

Regulatory Uncertainty

Lack of clear legal frameworks creates uncertainty for businesses and users.

- **Inconsistent regulations** across jurisdictions
- **Unclear tax** treatment and legal status
- Risk of restrictive policies



Governments and industry working toward balanced regulatory frameworks.

- **Regulatory Clarity:** EU, US frameworks (MiCA law - 2023)
- **Industry Standards:** Best practices emerging for compliance
- **Institutional Adoption:** Mainstream acceptance increasing

Emerging

Scalability

Current blockchains process fewer transactions than traditional systems.

- Bitcoin: **~7 transactions/sec** (Visa: ~65k/sec)
- High **fees** during network **congestion**
- Limits mass adoption for everyday payments



Multiple scaling solutions at various stages of implementation.

- **Layer 2:** Lightning Network, Rollups (1000s tx/sec)
- **Sharding:** Parallel processing (Ethereum 2.0)
- **New Consensus:** New mechanisms improving throughput

Active Development

The Programmable Trust: Four Key Takeaways

Essential concepts that define blockchain technology

1

Decentralized Ledger

Blockchain is a distributed database replicated across many nodes. No single entity controls it. Everyone has a copy, and everyone can verify the data.

2

Cryptographic Security

Blockchain uses cryptography to prove ownership and authenticity. Your private key is mathematical proof that you own your assets. No one can forge this proof.

3

Consensus Mechanisms

Blockchain achieves agreement without intermediaries through consensus mechanisms like Proof of Work or Proof of Stake. The network collectively validates transactions.

4

Transformative Potential

Blockchain can transform industries by enabling trust without intermediaries. From supply chains to finance to identity, the applications are vast and still being discovered.

"Blockchain is not a technology that will replace everything. It's a technology that solves specific problems: creating trust without intermediaries, ensuring transparency, and enabling programmable systems. When applied correctly, it has the power to transform how we organize society."

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Thanks for your time
Questions?
